P395: Guide 8: Monte Carlo Methods

1. Start up a new notebook and title it 'Guide8_your_name'. Enter
```
import matplotlib.pyplot as plt
import numpy as np
```

Introduction

In Activity Guide 2 on numerical integration, we saw that random sampling (a.k.a., a Monte Carlo method) could be used to evaluate high-dimensional integrals. Such high-dimensional integrals are commonplace in statistical mechanics where we compute macroscopic properties of a physical system in terms of averages over its microscopic degrees of freedom. The physical system can be summarized by a state vector $x \in R^N$ that assigns values to all *N* degrees of freedom (e.g., positions, momenta, angular momenta) of the microscopic system. Thus the dimensionality of the state space can be enormous. Each microscopic state *x* has a probability $P(x)$ for being observed. For some property (e.g., magnetization) that depends on the state of the system, $f(x)$, we can calculate its average through the integral

$$\langle f \rangle = \int \mathrm{d}x\, f(x)\, P(x)$$

Since the dimensionality of our state space is large, this integral is intractable using grid-based numerical-integration methods. We could resort to naive Monte Carlo that just picks states randomly, but we would waste a lot of time randomly sampling states x that have $P(\mathrm{x}) \approx 0$. In this guide we will look at a method that implements *importance sampling* so that we spend most of the time in regions where the probability is non-zero. Such Monte Carlo simulations are used in many areas of science, from studying the statistics of detectors in high-energy physics, to calculating the properties of materials in condensed matter, to studying information content in data science.

Metropolis-Hastings Monte Carlo Method

For systems in thermal equilibrium, statistical mechanics shows that $P(x)$ is the Boltzmann distribution. In short, when a system is in equilibrium at temperature T, the chance of observing a microscopic state x that has energy $E(x) = E$ goes as $\mathrm{P}(E) \propto e^{-\beta \mathrm{E}}$ where $\beta = 1/(k_\mathrm{B}T)$. The exponential factor is known as the Boltzmann weight. If two states from the system have respective energies E_1 and E_2, the ratio of their probabilities is

$$\frac{P_1}{P_2} = e^{-\beta(E_1 - E_2)} = e^{-\beta \Delta E}.$$

So, the chance of observing one state versus the other at equilibrium depends only on the energy difference between them.

For the integral above, we could calculate it as a simple average

$$\langle f \rangle = \frac{1}{N_\alpha} \sum_{\alpha=1}^{N_\alpha} f(x_\alpha)$$

if we could generate an ensemble of N_α states $\{x_\alpha\}$ whose frequency of occurrence matched $P(x_\alpha) = P\big(E(x_\alpha)\big)$ from the Boltzmann distribution. How do we sample an ensemble that satisfies the Boltzmann distribution?

There are several algorithms that generate an ensemble of states that satisfies the Boltzmann distribution. The most widely used is the Metropolis-Hastings algorithm. Instead of picking states purely randomly, it generates a new state in a random way that depends on the previous state. For example, consider a particle getting kicked around randomly on a bumpy landscape. When it is on top of a hill it has a similar chance of going left or right, but when it is on a slope it is more likely to go down than up. So how it moves depends on where it is located. In mathematics, a state/history-dependent stochastic process is known as a Markov process, and the sequence of randomly generated states is known as a Markov chain. It can be shown that on long time scales the resulting ensemble of states satisfies the desired distribution (in our case, Boltzmann).

The Metropolis-Hastings algorithm is as follows:

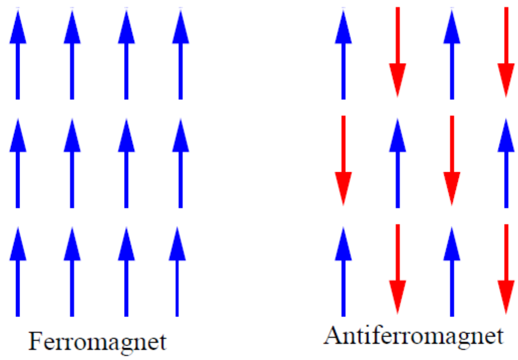
1. Start with some initial state configuration x, with energy $E(x)$
2. Make a move that changes the configuration to x', with energy $E(x')$
3. Accept the move with probability
$$P = 1, \qquad \Delta E = E(x') - E(x) \leq 0$$
$$P = e^{-\beta \Delta E}, \quad \Delta E = E(x') - E(x) > 0$$
4. If the move is rejected, set the state back to x. Otherwise the new state is x'.
5. Repeat steps 2-4 until equilibrium is achieved and you sample enough states to compute the desired averages.

Moves that lower energy are always accepted and moves that raise energy are rejected or accepted based on the Boltzmann weight. Physically, the picture is that thermal kicks can occasionally raise you to higher energy, and this is more likely at higher temperature.

The algorithm is straightforward and under the appropriate choice of moves will generate an ensemble of states that satisfies the Boltzmann distribution. The key word here is appropriate. Moves must be such that the system can explore all its phase space (if not, you won't get an equilibrium ensemble). The other big consideration is how to select states to include in your ensemble. One usually starts with a very unlikely initial configuration at the given temperature, so you must wait a certain number of moves until equilibrium is reached before you start sampling. And then because nearby states are correlated, you should only sample states every so often so that the states in the ensemble are not (strongly) correlated. Depending on the system, this sampling period could be very long (e.g., if the system has strong interactions or is near a phase transition when the system displays critical slowing down).

Ising Model

We will now use the Metropolis-Hastings Monte Carlo method to study one of the most famous models in physics, namely the Ising model. It was developed to explain the observed properties of magnetic materials. A ferromagnet, such as iron, develops a permanent macroscopic magnetization when its temperature drops below the *Curie temperature* T_C (1043 K for iron). Above this temperature it has no magnetization. Microscopically, a magnetic material is composed of many little magnetic dipoles. If they are randomly oriented then there is roughly no net magnetization, but if they align, then the material is magnetized.



The Ising model puts these dipoles, or 'spins' on a lattice, and instead of allowing them to point in any orientation, it assumes they can only point in two directions, either up or down. The spin on a lattice site can be described mathematically by a variable $S_i = \pm 1$, positive for up and negative for down. In a magnet, the microscopic dipoles interact magnetically such that it is energetically favourable for them to align. This interaction depends on the size of the dipoles, their relative orientation, and the coupling strength. In the Ising model, the interaction between two of these dipoles is simplified to be $-JS_iS_j$ for coupling energy J. The sign of *J* depends on the type of material. For ferromagnets, $J > 0$ and favours the spins being aligned (hence magnetized) whereas *J*<0 favours the spins being anti-aligned, leading to anti-ferromagnetic behaviour (shown on the right above). We will consider only the case of a ferromagnet where $J > 0$.

The Ising model makes the further restriction that <u>only the nearest-neighbour spins interact</u>, so the total energy of a particular configuration of spins on the lattice is

$$E = -\frac{J}{2}\sum_i \sum_{j \in \text{n.n.}} S_i S_j$$

For a square lattice, there are 4 nearest neighbours per site. The factor of ½ accounts for double counting each interaction over the two sums. This energy is used in the Metropolis-Hastings algorithm to determine if moves are accepted or rejected.

Calculating observables

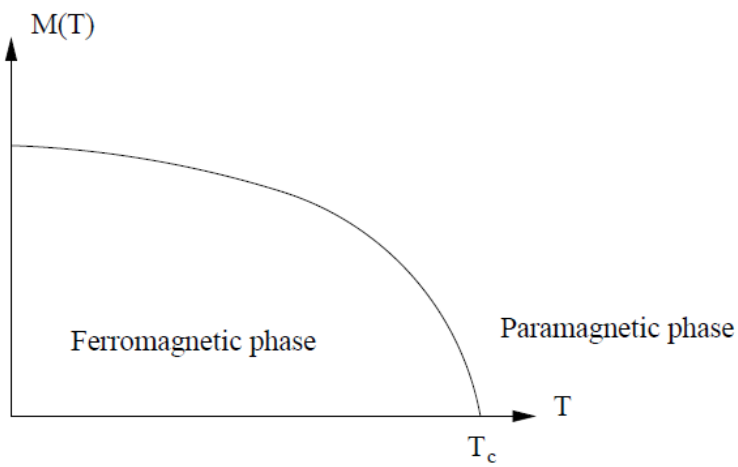
When the system of spins is in the particular configuration (state) α, the net magnetization is

$$M(\alpha) = \sum_i S_i$$

Averaging over an ensemble of N_α states generated at a particular temperature *T* gives the average magnetization

$$\langle M(T)\rangle = \frac{1}{N_\alpha}\sum_\alpha M(\alpha)$$

This quantity undergoes a phase transition at T_C, as shown in the figure below.



The 2D Ising model on a square lattice shows a 2nd-order phase transition (i.e., with discontinuous second derivative) for the case J>0 (i.e., a ferromagnet), and Lars Onsager solved the model exactly, showing that the critical temperature is

$$\boxed{k_B T_c \cong 2.27\ \mathrm{J}}$$

Besides the average magnetization, another macroscopic quantity of interest is the specific heat at constant volume

$$C_V = \left(\frac{\partial\langle E\rangle}{\partial T}\right)_T$$

In your statistical mechanics course, this derivative of the average energy with temperature can be related to the energy fluctuations at the given temperature as

$$C_V(T) = \frac{\partial\langle E\rangle}{\partial T} = \frac{1}{k_B T^2}\left[\langle E^2\rangle - \langle E\rangle^2\right]$$

So using our simulation at fixed T, if we calculate E and E^2 for each configuration we can then calculate the ensemble averages and find the specific heat.

Another related quantity is the magnetic susceptibility

$$\chi_M = \frac{\partial\langle M\rangle}{\partial H}$$

where H is the magnetic field, which can be related to the magnetization fluctuations at a given temperature:

$$\chi_{\mathrm{M}} = \frac{1}{k_{\mathrm{B}}T}[\langle M^2\rangle - \langle M\rangle^2]$$

Finite-size effects and scaling

One last issue: our simulations are on a finite-sized square lattice of *L* x *L* points. Right at a phase transition, the various phases exist at all scales (i.e., the arrangement of spins is fractal),so there is no well-defined length scale (we say that the correlation length diverges). When we do a simulation on a finite-sized system of length scale *L*, we artificially truncate this divergence at *L*. The result is that we smooth out the divergence at the phase transition, essentially destroying the phase transition. The result is that the above physical quantities now become dependent on our choice of system size *L*. Fortunately, we can work out how these quantities should scale with *T* and with size *L*, so we can still figure out where the true phase transition occurs. The result is that to explore the phase transition, we need to carry out simulations at several different system sizes (the bigger, the better) and quantify how the above quantities (i.e., magnetization, specific heat, susceptibility) vary with *T* and *L*.

Monte Carlo simulation of Ising Model

Here we will use the Metropolis-Hastings algorithm to simulate a finite-sized Ising model on an *L* x *L* square lattice. We will use energy units where $J = 1$.

Task 1: Pre-simulation Analysis

1. Analysis: In the Metropolis-Hastings algorithm, you need to calculate the energy change ΔE for flipping a particular site, $S_i \to -S_i$. Using the above formula for the Ising-model energy, work out a formula for ΔE. You should find that it only depends on the current value for the spin of the site that got flipped and its 4 nearest neighbours. This greatly speeds up the calculation, eliminating the need to do the full sum over all the lattice sites.

Periodic boundary conditions: You will apply periodic boundary conditions to your lattice. This means that sites on the top, bottom, and sides interact with each other (i.e., the left-edge spins are nearest neighbours to spins on the right edge). The calculation of ΔE requires knowing the four nearest neighbours of a given site.

2. By using some numbers for indices within the lattice and on the edges, convince yourself that the following equations give the correct indexes for the four neighbours for site (*i*,*j*) and also properly take into account the periodic boundary conditions:

```
left   = (i, j - 1)
right = (i, (j + 1) % L)
top    = (i - 1, j)
bottom = ((i + 1) % L, j)
```

Task 2: Metropolis-Hastings Simulation Code

Write code to simulate the Ising model on a square lattice of size *L* x *L* and at a given thermal energy $k_B T = 1/\beta$. Your code should do the following:

- Initialization: Initialize your *L* x *L* lattice with random +/- 1 spin orientations.
- Metropolis-Hastings sweeps: Carry out a user-specified number of 'sweeps', where each sweep consists of going through all $N = L^2$ sites, flipping each one, and applying the Metropolis-Hastings rule on whether to accept the move or not based on ΔE and the specified thermal energy $k_B T$. Use your pre-simulation results for computing ΔE and applying periodic boundary conditions. **TIP**: For each sweep, instead of going through each site in order, it is better to pick *N* lattice sites at random.
- Equilibration: Do a user-specified number of equilibration sweeps before you start sampling states for your ensemble.
- Sampling: After the equilibration sweeps have been completed, start sampling spin configurations at a user-specified interval of sweeps. For each sampled state of spins, compute and store the values of $|M| = |\sum_i S_i|$, $M^2 = (\sum_i S_i)^2$, and total energy *E* of the system (not ΔE).
- File output: You don't want to lose the data from a long run. Your simulation should output to a file the above computed values for each sample configuration. The result is a file with one row for each sample and one column for each computed variable in (4). Come up with a reasonable file-naming convention so that you know *L* and $k_B T$. (You might also consider outputting the spin configurations of each sample if you wanted to visualize what your system was doing – this can be helpful when debugging more complex models.)

Task 3: Data Generation

1. Testing and debugging: Adjust the output of your code so that it returns a list of arrays of the spin configurations of each sample. Run your code at i) high temperature ($k_B T$=3.0) and ii) low temperature ($k_B T$=1.5) and use `plt.imshow` to look at several of your sampled spin configurations at the two temperatures. At high temperature, the net magnetization of the lattice should be close to zero (i.e., an equal number of up and down spins). At low temperature, the net magnetization should be non-zero. Do your spin configurations agree with this? And can you tell if your periodic boundary conditions are being applied correctly?
2. Adjust your code so that it now outputs to a file the required calculated quantities specified in Activity 2. Run your simulation code for the following temperatures $k_B T$ = [1.5, 2.0, 2.1, 2.2, 2.25, 2.26, 2.27, 2.28, 2.29, 2.3, 2.5, 3.0] and system sizes, *L* = [10, 15, 20, 25]. Use 200 equilibration sweeps. Sample every 50 sweeps, and take 300 samples for the calculation of your ensemble averages below. (If you have time and are patient, you could increase both the sampling frequency and the number of samples taken).

Task 4: Analyzing your simulated data

1. From your samples, calculate at each *L* and *T* the ensemble average values $\langle |M| \rangle$, $\langle M^2 \rangle$, $\langle M^4 \rangle$, $\langle E \rangle$, and $\langle E^2 \rangle$.

2. Plot the mean absolute magnetization per spin $\langle |M| \rangle / N$ (*N* is the number of lattice sites) as a function of temperature at each *L*.
3. Analysis: How do your plots of average absolute magnetization compare to the schematic figure above that depicts a true 2nd-order phase transition? What do you notice as you change *L*? (Note: you probably will have noisy plots and this is due to not enough samples and likely sampling too frequently – i.e., your sampled states are not uncorrelated, or independent.)
4. Calculate and plot the specific heat per spin
$$\frac{C_\mathrm{V}}{N k_\mathrm{B}} = \frac{1}{N} \frac{1}{(k_\mathrm{B} T)^2} (\langle E^2 \rangle - \langle E \rangle^2)$$
versus temperature, at each *L*.
5. Analysis: What do you notice about the specific heat around the critical temperature of $k_B T_C \approx 2.27$? And how does this change with system size *L*?
6. Calculate and plot the (magnetic) susceptibility per spin,
$$\frac{\chi_\mathrm{M}}{N} = \frac{1}{N} \frac{1}{k_\mathrm{B} T} (\langle M^2 \rangle - \langle |M| \rangle^2)$$
versus temperature, at each *L*. Again, what do you notice about the susceptibility around the critical temperature? And how does this change with system size *L*?

One thing you should have noticed is that the temperature dependence of the macroscopic quantities also depends on the system size *L*. A neat statistic known as the Binder cumulant

$$g(L, T) = \frac{3}{2} \left(1 - \frac{1}{3} \frac{\langle M^4 \rangle}{\langle M^2 \rangle^2} \right)$$

doesn't have any *L* dependence at the critical temperature. This quantity tends to 1 for $T < T_\mathrm{c}$ and 0 for $T > T_\mathrm{c}$. Because of the *L* independence at the critical temperature, the curves at different *L* all cross each other at T_c.

7. Calculate and plot the Binder cumulant as a function of $k_\mathrm{B}T$ at your various *L*, and estimate the critical temperature based on where the curves cross. It's not perfect … it would be a lot better if you could simulate larger *L*.

Task 5: Wrap up

1. Please provide any feedback on questions that you found difficult to understand because of the wording.
2. Please provide any other comments about this notebook.

Extra Challenge (Finite-size scaling analysis): (*not graded*)

If you have time, this is a good activity to work through. It deals with the important topic of scaling and how to extract the behaviour of an infinite system using simulations of a finite system.

We will be considering a system that undergoes a phase transition. Using the critical temperature, we define a rescaled temperature $t = T - T_\mathrm{c}$. Near the critical temperature ($t = 0$), a number of physical properties of the system diverge (you saw this as peaks in your calculations above), going as $1/t^{\mu}$ where μ is some critical exponent that characterizes the divergence and differs from one property to the next. These properties include the correlation length ξ (the distance over which spins are correlated), the specific heat C_{V}, and susceptibility χ_{M}. Near the critical temperature, the quantities go as

$$\xi \sim t^{-\nu}$$
$$C_{\mathrm{V}} \sim t^{-\alpha}$$
$$\chi_{\mathrm{M}} \sim t^{-\gamma}$$

We can write the divergence for susceptibility (or specific heat) in terms of the correlation length by eliminating *t*:

$$\chi_M \sim \xi^{\gamma/\nu}$$

Now what is the effect of studying a finite-sized system of size *L*? When the correlation length is much less than *L*, then the susceptibility should go as the above scaling. However, when the correlation length exceeds the system size *L*, then the correlation length gets cut off, and this artificially truncates the susceptibility to be $\chi_{\mathrm{M}} \sim L^{\gamma/\nu}$. We can summarize this scaling of susceptibility with *L* as

$$\chi_{\mathrm{M}} = \xi^{\gamma/\nu}\,\chi_0(L/\xi) \tag{A.1}$$

with

$$\begin{aligned} \chi_0(x) &= \text{const.}, \qquad & x \gg 1 \\ \chi_0(x) &= x^{\gamma/\nu}, \qquad & x \to 0 \end{aligned}$$

(If you substitute $x = L/\xi$, you recover the desired limiting behaviour from above.) The problem with (A.1) is that it still involves ξ, which we don't know. We can eliminate the correlation length by defining a new scaling function,

$$\tilde{\chi}_0(\mathrm{x}) \;=\; x^{-\gamma}\,\chi_0(x^{\nu})$$

Using this and $\xi^{-1/\nu} \sim t$ gives our final scaling relationship that is just in terms of our finite system size *L* and reduced temperature *t*:

$$\chi_{\mathrm{M}}(L,t) = L^{\gamma/\nu}\tilde{\chi}_0\left(L^{1/\nu}t\right) \tag{A.2}$$

This is the scaling equation for the magnetic susceptibility near the critical temperature in finite-sized systems. It's quite amazing. Our finite-size Monte Carlo simulations give values for $\chi_{\mathrm{M}}(L,t)$, and if we plot on the *y*-axis the quantity $\chi_{\mathrm{M}}/\left(L^{\gamma/\nu}\right)$ versus a rescaled *x*-axis of $L^{1/\nu}t$, then all our simulations should collapse onto a single curve, namely the scaling function $\tilde{\chi}_0$! Of course this collapsing of all our different simulations onto a single curve requires that we choose the correct critical exponents γ, ν, and critical temperature T_{c} (which enters through $t = T - T_{\mathrm{c}}$). So this specifies a method for finding the critical exponents and critical temperature of a system undergoing a phase transition: 1) carry out simulations of your system at different system sizes; 2) plot the above rescaled quantities and vary the exponents and critical temperature until they all collapse onto a single curve.

1. See if you can work out the missing steps in going from equation (A.1) to equation (A.2).
2. Take your calculated values of the susceptibility $\chi_{\mathrm{M}}(T,L)/N$ at different values of *T* and *L*. At a given *L*, make a scatter plot of the rescaled *y*-values $(\chi_{\mathrm{M}}(T,L)/N)/L^{\gamma/\nu}$ versus a rescaled *x*-

variable $(T - T_c)L^{1/\nu}$. Do this for each L, on the same graph. Use $T_c = 2.27$, $\nu = 1$, and $\gamma = 1.75$. You should see that all the points collapse onto a single curve.

3. Vary T_c by 20% in either direction and see that the points don't collapse anymore. Do the same variation on the critical exponents and convince yourself that other values don't collapse the points onto a single curve.

There are other ways to find critical exponents (such as real-space renormalization group), but this method shows explicitly how one can take into account the effect of finite sizes. Why do we care about critical exponents? It has been found that completely different microscopic systems can have phase transitions with the same critical exponents. This leads to the topic of universality, where the idea is that near a phase transition the underlying microscopic details ultimately don't matter.